Ensembling-mRBF-LSTM Framework for Prediction of Abnormal Traffic Flows

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***Abstract*—The prediction of abnormal traffic flows has always been a primary concern in traffic management. If the management unit can predict the occurrence of abnormalities, it can manage and control transportation in advance in order to avoid abnormal traffic flows and enhance the service quality. While previous researchers generally predicted abnormal conditions in traffic flows by means of a single time series prediction model, such methods might result in inaccuracy in the prediction of abnormal traffic flows in some areas. As a result, in terms of practicality, the traditional methods fail to produce satisfactory results. Moreover, with the traditional methods, researchers often find it difficult to obtain the best time-delay item in the model and can only do so by using the trial-and-error method, which is nevertheless cost-prohibitive. In order to solve this problem, in this study, we predicted abnormal traffic flows by using the Ensembling-mRBF-LSTM framework, which consists of three new concepts: (1) introducing the concept of ensembling learning to the target issue and integrating the results produced by multiple existing prediction methods into the prediction of abnormalities, (2) devising the concept of mRBF-LSTM to select the prediction methods applicable to the prediction of abnormalities in this study from the existing methods and to decide on the applicable time-delay item for each existing prediction method, and (3) predicting abnormalities by using only the existing methods selected via mRBF-LSTM. With the aforementioned three major concepts, in this study, we sought to vastly improve the existing method for abnormal traffic flow prediction. Lastly, this study provided the traffic flow data of Taipei Mass Rapid Transit (MRT) in Taiwan to verify the effectiveness of the proposed method.**

Keywords—Abnormal Traffic Flows, mRBF, LSTM, Model Ensemble, Deep Learning

I. Introduction

The prediction of abnormal traffic flows has always been a primary concern of the government, because if the government can detect abnormal traffic flows in a certain area several hours in advance, it can tackle the problem beforehand and minimize the influence of the abnormal traffic flows on the people and the city, thereby enhancing people's convenience. An example of the most common prediction of abnormal traffic flows is illustrated in Figure 1, in which the solid line and the dashed line represent the change in the outbound traffic flow of an MRT station A on two different days, respectively. The gray lines represent the upper and lower bounds of the change in the regular outbound flow of A assessed by the government according to previous data. First, the solid line suggests that no particular incident occurred around A on that day and that the outbound flow per hour corresponded to the yearly average. As a result, the solid line remains between the two gray lines all the while. In contrast, the dashed line suggests that there was a particular incident from 6 p.m. to 8 p.m. on that day around A, and therefore, the outbound flow of A began to increase from 3 p.m., exceeded the upper bound of the regular range at 5 p.m., and was again lower than the upper bound upon the completion of the incident at 9 p.m. In this example, the interval between 5 p.m. and 8 p.m. was reckoned by the government as the period in which the traffic flow was abnormal. The concern regarding the prediction of abnormal traffic flows was to enable the prediction system to notify the government of the upcoming abnormal outbound flow within the next few hours when the outbound flow began to increase (despite still within the regular range) at 3 p.m. on that day, as indicated by the dashed line.

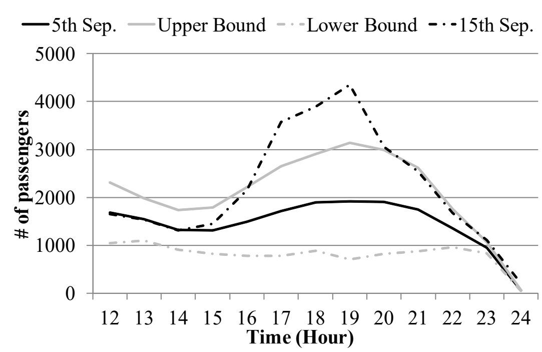

Figure 1. An example of abnormal traffic flows.

The previous concern regarding the analysis of the abnormal flows could be processed in three dimensions: (1) compiling statistics from the previous data before examining the position of the current data in the statistical model in order to identify abnormal conditions; such examples include box plot, three-sigma, and latent correlation probabilistic model [1]; (2) analyzing abnormalities by comparing the previous and the current data; such examples include *k*-nearest neighbors [2] and isolation forest [3][4]; and (3) inputting previous data into different types of time series

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modeling methods to construct a model, which will be utilized to detect abnormal flows according to the current or the future prediction results, with autoregressive integrated moving average models (ARIMA) [5][6], shallow recurrent neural networks (SRNN) [7][8], and long short-term memory (LSTM) [7][9] being the time series modeling methods commonly utilized based on this concept. In the three aforementioned concepts, the first two are not applicable to predicting abnormal traffic flows, as they only analyze the previous and the current data and cannot produce future data for analysis. In terms of the third concept, it enables us to predict future flows and abnormalities via a time series prediction model and is therefore applicable to the discussions in this study.

In general, during the application of the time series modeling method to all types of problems, two primary concerns should be taken into account: (1) the modeling method applicable to the time series to be modeled and (2) setting the time-delay parameters of the time series modeling method correctly. The first concern should be taken into consideration because while there are currently a great number of time series modeling methods, each method has its own advantages and deficiencies and produces different effects on the time series modeling results produced with the different characteristics of variation. For instance, although ARIMA [10] can lead to a certain degree of modeling accuracy for most time series, it can only capture the linear relations in the time series and cannot capture the non-linear ones, thereby producing unsatisfying modeling effects when it comes to the time series with non-linear variations. With respect to SRNN [11], although it can capture the non-linear variations in the time series by means of the non-linear activation functions in the neurons, it fails to capture the long-time connection in the time series and therefore produces poor results when applied to certain time series. Then, in terms of LSTM, although it can overcome the limit of SRNN and enhance the prediction accuracy by omitting previous and unimportant data, it produces unsatisfactory results when it comes to time series modeling with less variation due to the difficulty of optimizing the network structure and overfitting [12]. In view of the aforementioned situations, it is extremely difficult to select a suitable time series modeling method when the time series to be modeled remains unknown. In recent years, several researchers have proposed the notion of ensembling learning to tackle the aforementioned problems by combining multiple time series prediction results and generalizing an ultimate prediction result [13][14]. However, when making a generalization from various time series prediction results, taking the results produced by inapplicable methods into account altogether may result in lower prediction accuracy. When it comes to traffic flows, the selection of a time series model is a difficult task because of the diverse distinctions among the characteristics of variation of the traffic flow in each area. For instance, the outbound traffic flows of the MRT stations in the residential areas may be consistent and different from the model, while those in the business areas may be heavily influenced by the surrounding activities, and therefore are complicated and difficult to be modeled. In view of this, new methods have to be proposed to solve the problem concerning the time series models of traffic flows.

Another concern to be taken into account during the application of time series modeling is setting the time-delay parameters in the model. The time-delay parameters signify the previous time data utilized for future predictions. In consideration of the intrinsic periodicity of the time series, it is insufficient to simply take the first few time values into account for prediction, and a better prediction result can only be attained by gathering the flow values of the previous period. Two major methods have been proposed to assist the time series models in selecting the applicable time-delay parameters. For instance, the Lipschitz quotient criterion [15][16] finds the best time-delay parameters by examining the association of variation between the input and the output values under different time lags, while false nearest neighbors [17][18][19] does so by finding the nearest neighbor of the values output at different times. However, we discovered that it is extremely difficult to apply these two methods to the traffic flow issue. On the one hand, as the traffic flow data are generally gigantic, the two methods, which both utilize dual time points for the inspection of the relevance between the input and output values, would require a formidable amount of calculation time. On the other hand, when analyzing the outbound flow of the MRT stations, the best time-delay parameters for the residential and the business areas may differ and therefore need to be computed again. In view of these two concerns, we discovered that it is almost impossible to compute the time-delay parameters of all the MRT stations. Therefore, new methods are required to select the best time-delay parameter for modeling in order to further facilitate the process of time series modeling.

In this paper, we propose the idea of predicting abnormal traffic flows by using the Ensembling-mRBF-LSTM framework (EmRLF). The idea consists of three major concepts: (1) modeling and predicting the target traffic flows by using multiple existing time series modeling methods, and naming the time series obtained on the basis of the concept of the predicted time series of the existing methods (PTSs); (2) training a new deep learning model mRBF-LSTM, with the PTSs being the input and the target traffic flow the output, and disassembling the trained mRBF-LSTM in order to understand which of the PTSs can facilitate the modeling of the target traffic flow the most. The mRBF-LSTM discussed here is the reformed version of RBF-LSTM [20], which has been verified its ability to detect the most important input time series in the modeling process after being trained and disassembled. However, we found that because of a lack of concern for the time delay and the instability of the selection results of important inputs (different training and selection results), the structure of RBF-LSTM and its corresponding decomposition algorithm cannot be directly applied in this study. As a result, in this study, we reformed the structure and devised a new decomposition algorithm to solve the aforementioned problems, and named the reformed method modified RBF-LSTM (mRBF-LSTM); and (3) training another time series model of any kind after utilizing the important PTS of the traffic flow to be modeled using the trained and disassembled mRBF-LSTM, and lastly utilizing the output of the model to predict abnormalities according to the conditions defined by the users or the government.

It is expected that the three aforementioned methods, in comparison with the traditional methods, can attain several advantages: (1) overcoming the difficulty of applying a single time series prediction method to several time series with different characteristics of variation with the idea of ensembling learning; (2) avoiding inaccurate predictions resulting from the inapplicable methods taken into account, as seen in previous ensembling learning, by using the PTSs selected via mRBF-LSTM and most effective in modeling the target traffic flow; and (3) simultaneously enhancing the accuracy of predicting both the time series and the abnormalities by utilizing the ability of mRBF-LSTM to simultaneously select important PTSs for modeling and find the best time-delay parameters. Lastly, the study provided the authentic data of the traffic flows of Taipei MRT in Taiwan to verify the effectiveness of the proposed method.

The rest of this study is organized as follows. Section II introduces the related works of the study. Then, Section III offers an illustration of EmRLF, which is followed by the experimental simulations presented in Section IV. Finally, Section V discusses the conclusion and future work.

II. RELATED WORKS

This section introduces three common methods for analyzing abnormal time series: seasonal and trend decomposition using loess (STL), ARIMA, and classification and regression tree (CART).

A. STL

STL is an algorithm frequently used for analyzing abnormal time series. The algorithm can disassemble the time series data into the trend component, the seasonal component, and the remainder component via loess. Thus, STL can filter out the trend component and the seasonal component in the time series, and utilize the remainder component to detect abnormalities in the short term. Currently, STL is applied in various fields. For instance, Ho and Ting [21] utilized STL to detect abnormalities in the time series data of dengue fever. Duan *et al.* [22] used STL to detect abnormalities in the flow of usage of the base station. Fan *et al.* [23] used STL to disassemble the time series data of the load before conducting the load forecasting of a short-term distribution network, in order to reduce the influence of the abnormal values on the forecasting model.

B. ARIMA

ARIMA is a well-known time series prediction model frequently utilized for analyzing abnormal time series by comparing the prediction results produced by ARIMA with authentic data. Schmidt *et al.* [24] verified higher accuracy and a lower rate of false positives in the detection of abnormalities by ARIMA. Ray *et al.* [25] used ARIMA in an early warning concerning children's immunodeficiency lead testing and the pneumococcal vaccine. More recently, Diab *et al.* [26] utilizes ARIMA to detect abnormal network traffic.

C. CART

CART is another common method for detecting abnormal time series. It trains the model with an unsupervised learning algorithm and predicts the next data point of the time series data. Then, it computes the error between the prediction result it produces and the actual data, and finally checks whether there is any abnormality in the data by using the extreme studentized deviate test. Kerdprasop *et al.* [27] used CART to detect abnormalities in the traffic of the clients' orders. Aung and Min [28] utilized CART to detect abnormalities in network traffic in order to prevent cyberattacks. More recently, Kanev *et al.* [29] used CART to detect abnormalities in the sensor of the smart home system.

III. ENSEMBLING-MRBF-LSTM FRAMEWORK

The Ensembling-mRBF-LSTM framework (EmRLF) proposed in this paper consists of two rounds of actions, as illustrated in Figure 2. The first round consists of three procedures: (1) training multiple existing time series models using actual traffic values and recording all the PTSs; (2) training the mRBF-LSTM model with the PTSs being the input and the target traffic flow value the output; and (3) disassembling the parameters within mRBF-LSTM to find the most important PTSs for the target traffic flow. The second round consists of two procedures: (1) training a time series model of any kind using the important PTSs identified by mRBF-LSTM as the input and the target traffic flow data as the output; (2) examining whether any abnormality occurs in the predicted values produced during the previous procedure according to the conditions provided by the user. The following part of this paper offers a detailed demonstration of EmRLF; however, considering the page limit, we only introduce the two most crucial procedures: (1) modeling of mRBF-LSTM (i.e., the second procedure of the first round) and disassembling mRBF-LSTM to obtain the most important PTSs in modeling the target traffic flow (i.e., the third procedure of the first round).

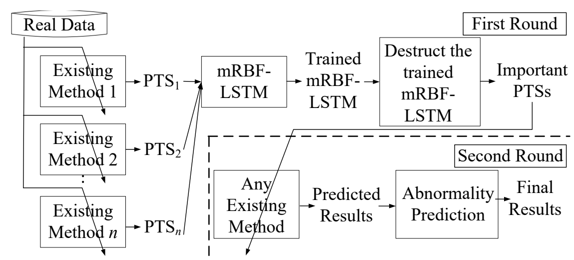


Figure 2. The flow chart of Ensembling-mRBF-LSTM framework.

A. Introduction of the mRBF-LSTM model

This section introduces the structure of the mRBF-LSTM model (as shown in Figure 3) and the formula for the structure of each layer: the input layer, the delay layer, the RBF layer, the recurrent layer, and the fully connected layer. The following introduces the content of each of these layers.

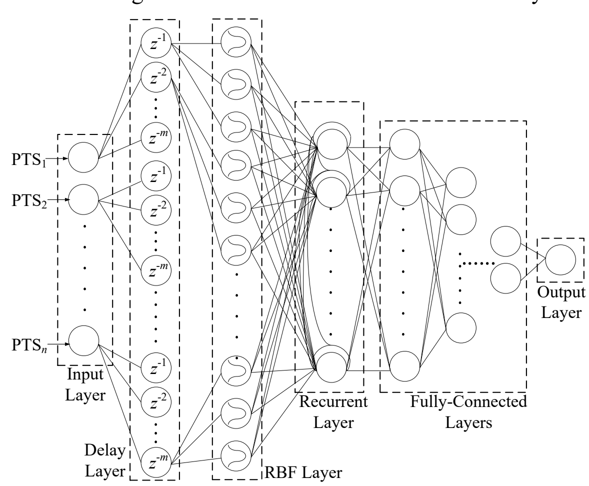


Figure 3. The Structure of mRBF-LSTM.

Input of mRBF-LSTM: Assume that in the first procedure of the first round, EmRLF selects n time series modeling methods to model actual traffic flows. We would then expect to obtain n PTSs, which would all be input into mRBF-LSTM.

Input layer of mRBF-LSTM: As there are n time series inputs in mRBF-LSTM, the number of neurons in this layer would also be n. In addition, the neurons in this layer would not be computed and would only deliver the model inputs to the delay layer.

Delay layer of mRBF-LSTM: The layer displays the most distinct difference between the structure of mRBF-LSTM and that of RBF-LSTM devised by [20]. The layer aims to disassemble each input PTS into multiple time series with a different reference time considered. Assume that a PTS originally input into mRBF-LSTM is the prediction result of a time point t produced by a previous time series model. The delay layer will then be able to rebuild the result of the PTS in $t-1$, $t-2$, ... The PTS disassembled into multiple time references will be regarded as a new input time series. The networks that follow will select from among all the time references in the PTS the most important ones for the target traffic flow model, which will entirely correspond to the previous concept of the time delay choice.

Assume that there are m time-delay items to be considered; that is, we want to find the most ideal time-delay item for the target traffic flow from among $t-1$, $t-2$, ..., $t-m$. The layer will then use $n \times m$ neurons. Each neuron in the input layer will connect m neurons, and the output of the (i, j)-th neuron in the delay layer $O_{(i,j)}^{delay}$ can be written in the following form:

$$O_{(i,j)}^{delay} = z^{-j}\left(O_i^{input}\right) \tag{1}$$

While i represents the i-th neuron in the input layer, j represents the j-th neuron in the delay layer corresponding to the i-th neuron in the input layer, and $z^{-j}(\bullet)$ represents j time points delayed, and O_i^{input} represents the output of the i-th neuron in the input layer.

RBF layer of mRBF-LSTM: Each neuron in the RBF layer uses the RBF function as the activation function. Such a design is affected by the notion, previously proposed by the radial basis function neural network [30][31], fuzzy neural network [32][33], and RBF-LSTM [20], that connects the RBF layer before the neural networks, and training them together will enable the RBF layer to mark the outputs from the outputs of the previous layer that can facilitate the succeeding network modeling. Assume that we will use k RBF functions to represent the output of each neuron in the delay layer. The layer will then have $n \times m \times k$ neurons. The (i, j, k)-th neuron in this layer can be written in the following form:

$$O_{(i,j,k)}^{\mathrm{RBF}} = \exp\left[-\frac{\left(O_{(i,j)}^{\mathrm{delay}} - m_{(i,j,k)}\right)^2}{\left(\sigma_{(i,j,k)}\right)^2}\right], \tag{2}$$

Eq. (2) is a Gaussian function, while $m_{(i,j,k)}$ and $\sigma_{(i,j,k)}$ represent the average value and the standard deviation of the Gaussian function, respectively.

Recurrent layer of mRBF-LSTM: This layer primarily accepts the output of the RBF layer and examines the influence of the output of this layer at a previous time point at that moment. The number of neurons in this layer, $floor((n \times m \times k)*0.5)$, is based on that in the previous layer; here, $floor(\bullet)$ signifies the removal of the digits after the decimal point. In addition, all the recurrent items between the neurons in this layer are in a fully connected state and can be written in the following forms:

$$U^t = I^t \times w_i^t + bis_i, \tag{3}$$

$$S^t = h_i^t \times kw_i^t. \tag{4}$$

$$h_i^t = O_i^{t-1}. \tag{5}$$

$$O_i^t = act\left(U^t + S^t\right). \tag{6}$$

Here, U^t represents the input equation at the time point t and is obtained by multiplying the current input I^t by the corresponding weight w^t and adding the corresponding $bias_i$. S^t is obtained by multiplying the output of the previous time point (i.e., $h^t_i = O^{t-1}_i$) by the corresponding kernel weight kw^t_i. Finally, the output of the current time point O^t_i can be obtained by inserting the sum of the two aforementioned values in the activation function $act(\bullet)$.

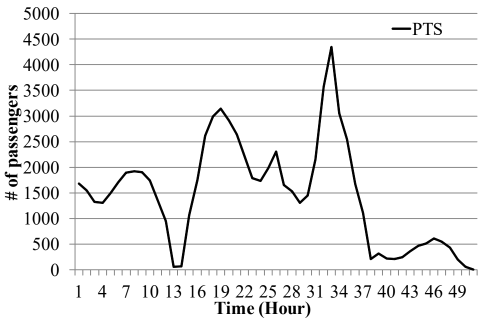
Fully connected layer of mRBF-LSTM: There may be multiple fully connected layers in mRBF-LSTM. Their primary object is to accept the output of the recurrent layer and output the final prediction value. The number of neurons in each fully connected layer hinges on that in the previous layer. Assume that there are α neurons in the previous layer. The succeeding fully connected layer will then have $floor(\log_\delta \alpha)$ layers, with δ being the parameter designated by the user. When $floor(\log_\delta \alpha)$ eventually equals 1, the corresponding fully connected layer will be the final layer of mRBF-LSTM. The i-th neuron in each layer can be written in the following form:

$$o_i^{fully} = act\left(i_i^{fully} \times w_{ij}\right) + b_i, \tag{7}$$

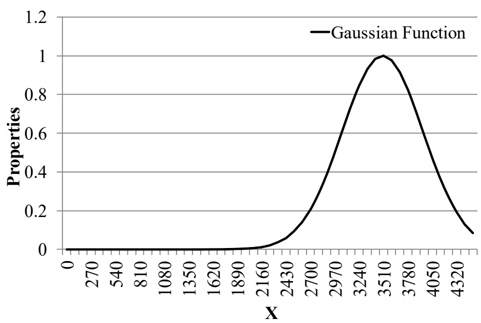
while o_i^{fully} represents the output of the neuron in the layer, $act(\bullet)$ represents the activation function, and w_{ij} represents the weight of the i-th neuron in this layer and the j-th neuron in the previous layer. Finally, b_i represents the bias of the neuron.

B. Disassembling mRBF-LSTM to select PTSs most effective in modeling target traffic flow

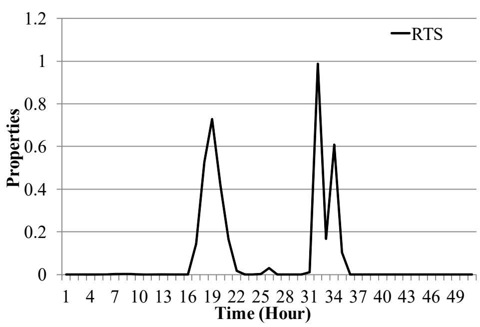
A disassembling method had been devised by [20] to help select the most important input in RBF-LSTM modeling. However, we found that the method devised in the early study selects a different input each time. This problem occurs because, irrespective of the method's concept that a multi-round selection process can enhance the precision of the selection results, the selection process in each round is conducted according to only a single training result of RBF-LSTM, and different selection results may be produced because of the different initial random parameters of the model. In order to solve this problem, in this paper, we propose a new selection mechanism, using only one round, in which multiple RBF-LSTM training results can be selected from. Thus, we expect to solve the problem caused by different selection results produced by the different initial random parameters of the model, as shown in [20], and facilitate the succeeding modeling procedures with the fixed selection results. The method consists of three procedures: (1) training mRBF-LSTM repeatedly for β times and acquiring β trained mRBF-LSTM; (2) collecting the RBF neuron output time series of each time-delay item of each PTS in these β mRBF-LSTM and checking whether each output time series affects the output of mRBF-LSTM; and (3) following the results produced in the second procedure and arranging the time-delay items of each PTS according to their degree of importance in mRBF-LSTM modeling.



(a)



(b)

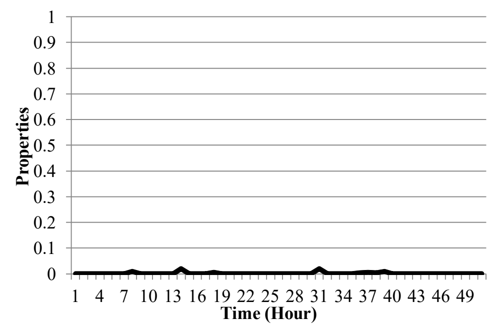


(c)

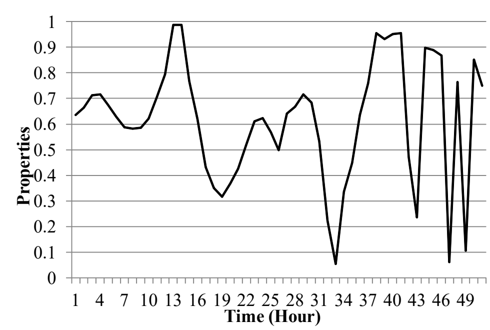
Figure 4. An example of the PTS, the corresponding RBF function, and the RTS. (a) PTS, (b) RBF function with m=3500, σ=450, (c) RTS

After training β mRBF-LSTM in the first procedure, in the second procedure, we collected the RBF neuron output time series of each time-delay item considered in each PTS, as illustrated in Figure 4. Assume that in each mRBF-LSTM, we use n PTSs in total, and for each PTS, m time-delay items are considered, each of which is represented by k RBF. For each time-delay item of each PTS, we then collect $\beta \times k$ RBF neuron outputs. In the model, $n \times m$ PTS time-delay items need to be considered. For the sake of convenience in the following explanation, we have abbreviated each RBF neuron output time series to "RTS."

After the completion of the aforementioned procedures, we determined whether each RTS would influence the output of mRBF-LSTM and keep the influential ones. Figure 5 demonstrates the core concept of and the method for the determination process. The line in each subfigure of Figure 5 represents an RTS. For the RTS in Figure 5(a), the value remained almost 0 at any time point. In this case, the RTS Was ineffective in modeling as there was no variation in the output value even after the RTS was inserted in the succeeding recurrent layer and fully connected layer. In Figure 5(b), the RTS vastly changed over time. In this case, the RTS was important for the training of the model because it affected the output of mRBF-LSTM to a certain extent. On the basis of this concept, we devised a method to examine whether the RTS would influence the output of mRBF-LSTM. First, we calculated η, which had the largest proportion of the same value in the RTS, and set $\eta\_threshold$ as the threshold. Whenη > $\eta\_threshold$, the RTS had no effect on the output of mRBF-LSTM. In contrast, when η < $\eta\_threshold$, the RTS affected the output of mRBF-LSTM.



(a)



(b)

Figure 5. Examples of RTSs. (a) the RTS remained almost 0 at any time point. (b), the RTS vastly changed over time.

The objective of the third procedure was to arrange the time-delay items of each PTS according to their degree of importance in mRBF-LSTM modeling by following the results produced in the second procedure. Assume that each time-delay item of the PTS obtained $\beta \times k$ RTSs after the second procedure, and the λ of these RTSs would affect the output of mRBF-LSTM. We then calculated the value of $\lambda/(\beta \times k)$. The higher the value was, the more effective was the time-delay item of the PTS in modeling the target output. The user could select from the top results arranged for the input of any type of the time series model in the succeeding round.

IV. SIMULATIONS

This section provides the traffic flow of each Taipei MRT station in Taiwan to verify the effectiveness of the proposed method. The section consists of three sections: (1) introduction of the dataset, (2) introduction of the experimental parameters and the experimental control groups, and (3) comparison of the effectiveness of the proposed method and that of the control groups. All the experiments in this study were conducted with Python on a Windows 10 system, with Intel Core I7-6700 CPU at 3.40 GHz, Nvidia GTX 1070ti GPU, and 64-GB memory.

A. Dataset of the traffic flow of Taipei MRT in Taiwan

In this study, we chose Taipei MRT in Taiwan for the examination of the prediction of abnormal traffic flows because Taipei MRT is applicable to the discussions on abnormal conditions, considering the accumulated operating time of its system (a period of 24 years starting from 1996 to 2019), the passengers' fixed commuting patterns, and the rather stable flow variation of the system. In addition, the yearly traffic flow of the system in 2019 was 789.6M, with the average daily traffic flow reaching 2.16M. In consideration of this massive number, the abnormal conditions would not be affected by small-scale emergencies, which Was another reason for selecting this dataset for this study.

According to the dataset, we collected the data of the outbound flow of each MRT station, from January 1, 2019, 12 a.m., to December 31, 2019, 11 p.m. Then, from all the 131 MRT stations, we selected five MRT stations for the analysis: Taipei Main Station (TR), Songjiang Nanjing (SJ), Taipei 101 (101), Dingxi (DX), and Xinpu (XP), with Taipei Main Station being the most crucial transfer station in the system and having the highest traffic volume. Both Songjiang Nanjing and Taipei 101 are located in major business areas, while Dingxi and Xinpu are both major residential areas. The average number of passengers per hour, the maximum number of passengers, and the standard deviation of all the times of the five stations are shown in Table 1.

Table 1. Statistics on the number of passengers at each station

Station	Avg.	Max	STD	Threshold of Abnormality
TR	7975	28067	4743.4	17461.8
SJ	1826	14266	2354.6	6535.2
101	1779	13191	1529.9	4838.8
DX	1854	7077	1487.6	4829.2
XP	1980	7793	1532	5044

B. Introduction of the experimental parameters and the experimental control groups

The internal parameter setting of EmRLF in the study was as follows: (1) using the prediction results produced by the three most common methods for analyzing time series abnormalities, ARIMA, CART, and STL, as the inputs of mRBF-LSTM; (2) consideration to the delay of three time points in each delay layer of mRBF-LSTM, with three RBFs in each RBF layer utilized to represent the output of each delay layer (a method proposed in [20]); (3) in terms of the selection of the important input of mRBF-LSTM, the $\eta\_threshold$ was set to 0.2; (4) in terms of the second round in EmRLF, LSTM, the best time series model generally recognized currently, was used for modeling; and (5) in terms of the definition of the abnormality of the traffic flow of the MRT station, we used the most general method and only considered it abnormal when the traffic flow of an MRT station exceeded the sum of the average number of passengers per hour of the corresponding MRT station in the past and a double standard deviation. Here, the triple standard deviation was excluded from consideration, because the cases where the traffic flow exceeded the triple standard deviation were statistically rare. The extremely small number of such abnormal conditions might not facilitate the experimental verification conducted in this study.

In order to verify the effectiveness of the proposed method, we selected six methods as the experimental control groups. The first three control groups discuss the use of a single method, using ARIMA, CART, STL, and LSTM to detect abnormalities. The other three control groups are devised on the basis of the concept of ensembling learning, and include the following: (1) detecting abnormalities by using the average value of ARIMA and CART; (2) using the prediction results of ARIMA, CART, and STL as the inputs of LSTM; and (3) using the parameters identical to those used in EmRLF in the study and replacing the function of mRBF-LSTM with a linear discriminant analysis (LDA).

LDA is applicable to this method because the analysis of abnormalities can also be regarded as a matter of classification, to which LDA is applicable. In addition, for the purpose of easy demonstration, we abbreviated the six control groups to "AR," "CA," "ST," "AV," "LS," and "LD," in that order, and the proposed method to "Our."

Two types of standards were used in this study to verify the effectiveness of the proposed method and that of the control groups, including root mean square error (RMSE), which was used to verify the time series prediction capability of all the methods, and accuracy, which was used to calculate the number of the actual abnormal conditions in the MRT traffic flow identified by each algorithm. The commonly used recall method, which examined the number of the actual abnormal conditions in the MRT traffic flow within the abnormal conditions identified by each method, will not be discussed considering the lack of attention given to it by the government or the traffic management unit.

C. Comparison of the effectiveness of EmRLF and of the control groups

Table 2 shows the comparison of RMSE of the proposed method and that of the five control groups. Note that STL was excluded from the discussion in this table considering that it was not a time series modeling algorithm. First, in terms of the use of a single method, the effects produced by AR were noticeably inferior to those produced by CA and LS, because while the traffic flows were mostly non-linear time series, AR produced poor modeling results when it came to non-linear time series. In terms of ensembling learning, the effects produced by AV noticeably fell between those produced by AR and CA, which attested to the fact that the modeling effects would be affected by the poor inputs if all the inputs were taken into account in ensembling learning.

Table 2. The RMSEs of EmRLF and six control groups

	Single Method		Ensembling Learning			
Station	AR	CA	AV	LS	LD	Our
TR	1955	1467	1382	1632	8379	**1075**

SJ	2146	365	1149	506	3312	**248**
101	1080	585	681	608	3677	**407**
DX	688	325	418	387	2588	**234**
XP	769	436	468	464	2072	**285**

With respect to LD, it produced poor modeling effects because the LDA method that it used was efficient in classification, but not in time series modeling. However, the time series prediction effects that the proposed method produces were evidently better than those produced by the other methods, even surpassing those produced by LS. This was because the proposed method first selected the prediction results most important for modeling the target traffic flow from among the results produced by ARIMA, CART, and STL, instead of using all of the prediction results for modeling. Thus, the poor prediction results were filtered out in ensembling learning, which led to higher modeling accuracy.

Table 3 shows the comparison of accuracy of the proposed method and of the six control groups. In terms of the use of a single method, we observed that AR produced better results in stations such as TR, DX, and XP, while the results it produced in SJ and 101 were inferior (Note that TR's accuracy is much worse than other stations, because TR is the largest station in the entire MRT system. The maximum traffic per hour is close to 30,000, and the standard deviation per hour is close to 5,000. These two values both are twice larger than other stations in the MRT system. Such drastic changes make TR more difficult to predict than other stations, so the prediction accuracy is lower.). With respect to CA and ST, both of them performed better in SJ, yet produced inferior results in the cases of the others. This corresponded to our observation regarding the discrepant qualities of the predictions of abnormalities of different stations when applying a single method. In terms of ensembling learning, although both AV and LS produced some of the best results in predicting the abnormalities of several stations, the performances of these two methods were unstable, which was likely to result from the poor prediction results input. However, when we used LD and the proposed method, the best prediction results could be achieved in terms of most of the stations, because in both of the methods, the inputs of little value to modeling were already filtered out before the second round of modeling. Nevertheless, when we further examined the performance of LD, we found that it still performed poorly in terms of some of the stations, because the LDA method that it used for the input selected the data on the basis of the classification. As a result, the selected inputs might not be applicable to the inputs required for the succeeding LSTM modeling process, hence reduced efficiency in prediction. With respect to the proposed method, as mRBF-LSTM selected the important inputs for modeling on the basis of time series, the best results were produced in terms of three of the five stations, with the results of the other two stations nearly reaching the optimal solution, which attested to the effectiveness of the proposed method.

Table 3. The accuracies of EmRLF and six control groups

Station	Single Method			Ensembling Learning			
	AR	CA	ST	AV	LS	LD	Our
TR	**0.66**	0.20	0.34	0.27	0.32	0.16	**0.66**
SJ	0	**1**	**1**	0.98	0.98	0.40	**1**
101	0.19	0.67	0.76	0.17	0.84	**0.94**	0.84
DX	0.97	0.51	0.57	0.98	0.96	**0.99**	0.96
XP	0.88	0.52	0.5	**1**	0.99	0.20	**1**

V. Conclusion

The prediction of abnormal traffic flows has always been a major concern in traffic management. While previous researchers often used a single algorithm for predicting abnormalities in traffic flows, we found that these methods might result in a discrepancy in the prediction accuracy when it came to abnormal traffic flows in different areas, which indicated the low applicability of these algorithms. On top of this, many researchers also found it difficult to obtain the best time-delay items for these prediction methods. In order to solve the two aforementioned problems, in this paper, we proposed EmRLF to predict abnormal traffic flows. With this new framework, we could overcome several problems that might occur when applying traditional methods by (1) using the concept of ensembling learning to overcome the challenge of applying a single time series prediction method to time series with different characteristics of variation; (2) selecting from all the prediction methods the ones that were the most important for the prediction of actual flows by using mRBF-LSTM before modeling, and thus achieving higher prediction accuracy in comparison with that produced with the traditional ensembling learning method, which simply combined the results produced by different methods; and (3) simultaneously finding the best time-delay parameters while selecting important prediction methods by mRBF-LSTM, and enhancing the time series prediction accuracy. Lastly, with this framework, we could provide more accurate predictions regarding traffic flows and the occurrence of abnormalities in the traffic flows. This was also verified by using the authentic traffic flow data of the Taipei MRT system in Taiwan.

In the future, we will combine the EmRLF with more anomaly detection methods, and explore whether these newly added anomaly detection methods can improve the anomaly prediction capabilities of EmRLF model.

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